

Convolutional and Recurrent Networks

Specialized structure for images and sequences

Neural Networks and Deep Learning Basics · Foundations of Machine Learning



Dense Layers Ignore Image Structure

Parameter explosion

224×224×3 pixels feeding 1,000 units means about 150 million weights.

Lost locality

Flattening does not tell a dense layer which pixels are neighbors.

CNN response

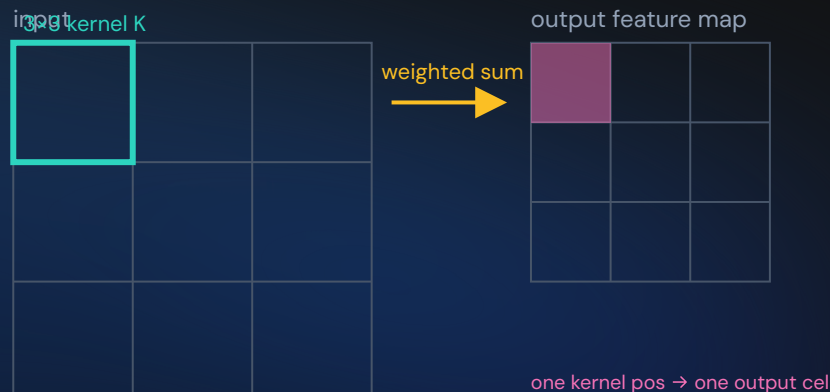
Share small filters across nearby spatial regions.

Convolution Shares a Local Filter

Small kernel — A learned 3×3 grid slides across the image.

Weight sharing — The same nine weights detect a pattern everywhere.

Feature maps — Different filters learn edges, textures, and shapes.



$$Y_{ij} = \sum_{u,v} K_{uv} X_{i+u,j+v}$$

A Compact CNN

Convolve + pool

Extract and downsample spatial features.

Flatten late

Preserve structure until learned features exist.

```
cnn = keras.Sequential([
    layers.Input(shape=(28, 28, 1)),
    layers.Conv2D(32, 3, activation="relu"),
    layers.MaxPooling2D(2),
    layers.Conv2D(64, 3, activation="relu"),
    layers.MaxPooling2D(2),
    layers.Flatten(),
    layers.Dense(10, activation="softmax"),
])
```

RNNs Share Weights Across Time

Ordered input — Text, time series, and audio depend on sequence.

Hidden state — Carry a learned summary from earlier steps.

Time sharing — Reuse the same transition weights at every position.

1

$x_1 \rightarrow h_1$

2

$x_2 \rightarrow h_2$

3

$x_3 \rightarrow h_3$

4

output

$$h_t = \sigma(W_{hh}h_{t-1} + W_{xh}x_t + b_h)$$

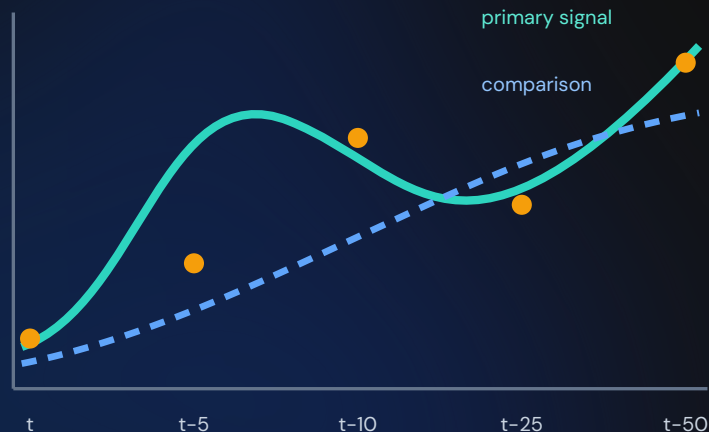
For the LSTM cell's gates and the full backpropagation-through-time walkthrough, see the **RNNs and LSTMs** deck.

Long Sequences Recreate Vanishing Gradients

Backpropagation through time — The chain rule multiplies through many recurrent steps.

Plain RNN — Long-range dependencies become hard to learn.

LSTM / GRU — Gates create paths that preserve information and gradient flow.



Choose Architecture From Data Structure

Dense

Tabular data with no special spatial or sequential structure.

CNN

Grid-like local structure: images and spectrograms.

RNN / LSTM

Ordered, variable-length sequences.

Beyond

Transformers increasingly dominate many sequence tasks.

Same Machinery, Specialized Layout

CNN

Share across space.

RNN

Share across time.

Both

Train with backpropagation.

Next: batch normalization and dropout make deep training more reliable.